

Module 05: Action in Robotics

Learning Objectives

1. Define **Action** within the context of Robotics.
2. Analyze how Action interacts with other components of the Active Inference framework.
3. Apply specific constraints of Robotics to the formal definition of Action.

Introduction

This module explores **Action**. In the **Robotics** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Action is a critical component of the 8-part Active Inference spine, bridging the gap between [Previous Topic] and [Next Topic].

Key Concepts

1. Action as a Markov Blanket Boundary

How does Action define the boundary between the agent and the environment?

2. Generative Models of Action

What parameters involved in Action must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Action drive the perception-action loop?

Applications

In Robotics, we see Action manifest in: * **Specific Example 1**: [Add domain-specific example here] * **Specific Example 2**: [Add domain-specific example here]

Conclusion

Understanding Action allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.